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Morning
01 JUN 2017

Uni. Roll No. ....

Program/ Course: B.Tech. (Sem. 4th)
Name of Subject: Computer Architecture and Microprocessor
Subject Code: IT-14405
Paper ID: 15536

Time Allowed: 03 Hours **Max. Marks: 60**

NOTE:

1) **Section-A** is **compulsory**
2) Attempt any **four** questions from **Section-B** and any **two** questions from **Section-C**
3) Any missing data may be assumed appropriately

Section – A [Marks: 02 each]

Q1.

a) Explain the difference between hardwired control and micro programmed control?

b) Explain the overflow condition in arithmetic shift micro operation?

c) How do you improve the cache performance?

d) Explain what is DMA?

e) Difference between RICS and CISC architecture?

f) What are the major difference between 8085 and 8086 Microprocessor?

g) Give two major applications of Microprocessors?

h) What are the two instructions needed in the basic computer in order to set E flip-flop to 1?

i) What is an instruction format?

j) What is asynchronous data transfer?

Section – B [Marks: 05 each]

Q2. Explain in detail RISC pipeline. Why is the cache miss penalty, greater in deeply pipelined processor?

Q3. Explain the circuit of Accumulator logic?

Q4. Explain memory hierarchy in a computer system?

Q5. Register A holds binary. Determine the register B operand and the logic micro operation to be performed in order to change the value (i) 01101101 (ii) 11111101

Q6. Why does DMA have priority over CPU when both when both request a memory transfer?

Section – C [Marks: 10 each (05 for each sub-part, if any)]

Q7.

a) How many types of address sequence are required in a control memory? Explain with the diagram of a control memory and its associated hardware.

b) With the help of flow chart, discuss the hardware divide algorithm. Explain how the divide overflow conditions are handled?

Q8. A computer has 16 register, an ALU with 32 operations and a shifter with eight operations all Connected to a common bus system.

(i) Formulate a control word for a micro operation.

(ii) Specify the number of pits in each field on the control word and give a general encoding scheme?

Q9. Write Short Note on

(a) Data Transfer Operations

(b) Address Sequencing

(c) Input/Output Interface

(d) D)Flip Flops

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